

AI Assignment 1 - System Test Log

Group Code: AI_A1_G08 | Generated: 2026-10-01 11:16:17

1. Verification Summary

Test Category	Command / Target	Expected Result	Status
Pipeline Execution	python run_all.py	Pipeline executes end-to-end	PASS
Unit & Integration	pytest	17 unit & integration tests pass	PASS
Artifact Generation	artifacts/ & models/	All JSON, PNG, joblib models present	PASS
Valid Inference	predict.py --record '{...}'	Returns structured JSON predictions	PASS
Invalid Inference	predict.py --record '{missing}'	Gracefully handles error with msg	PASS
Environment Cleanliness	clean .venv & dependencies	Requirements reproducible without cache	PASS

2. Artifact Verification

File Name	Expected Artifact	Verified Status
classification_metrics.json	Classification metrics (F1, Precision, Recall)	Verified
clustering_metrics.json	Silhouette & Davies-Bouldin Scores	Verified
regression_metrics.json	MSE, RMSE, R ² Scores	Verified
cluster_plot.png & confusion_matrix.png	Visualization PNG plots	Verified
classification_model.joblib	Serialized classification model	Verified
clustering_model.joblib & regression_model.joblib	Clustering & Regression models	Verified